# Power Supply Systems

## Part II

Phase 8: Fuse and Protection

**Aim:** To learn fuse, HRC fuse, relay, circuit breaker, isolator and basic protection from absolute zero.

**Style:** Simple English, SN ma'am style, clean diagrams, no overlapping words, every diagram word explained.

**Target:** Definitions, diagrams, working, comparison tables, protection flow, time-current idea and semester-ready short notes.

| | |
|---|---|
| **Subject:** | Power Supply Systems |
| **Section:** | Part II |
| **Phase:** | 8 – Fuse and Protection |
| **Level:** | Absolute zero to semester-question-ready |

Academic article style notes

# Contents

# 1 Phase 8 Roadmap

**Phase 8 Goal**

After studying distributors, we must protect them. Phase 8 explains how faults are detected and removed using fuse, HRC fuse, relay, circuit breaker and isolator.

**Phase 8 Learning Flow**

**Need of Protection**
↓
**Fuse and Fuse Terms**
↓
**HRC Fuse Construction and Working**
↓
**Relay and Circuit Breaker**
↓
**Isolator and Protection Sequence**
↓
**Comparison Tables and Numericals**

**How to Read the Diagram**

| Word | Meaning |
|---|---|
| Need of Protection | Why electrical equipment must be protected from faults and overloads. |
| Fuse and Fuse Terms | Basic protective device and important terms like rated current, fusing current, prospective current and pre-arcing time. |
| HRC Fuse | High Rupturing Capacity fuse used for safely interrupting high fault currents. |
| Relay and Circuit Breaker | Relay detects abnormal condition; circuit breaker opens the faulty circuit. |
| Isolator | Mechanical switch used for isolation after the circuit is de-energised. |
| Comparison Tables and Numericals | Semester-ready differences and small calculations. |

# 2 Need of Protection in Power System

**Definition**

**Protection** means the arrangement used to detect abnormal conditions and disconnect the faulty part from the healthy part of the power system.

Electrical faults may occur due to:

- short circuit,
- overload,
- insulation failure,
- earth fault,
- lightning,
- equipment failure,
- wrong operation.

Protection is needed to:

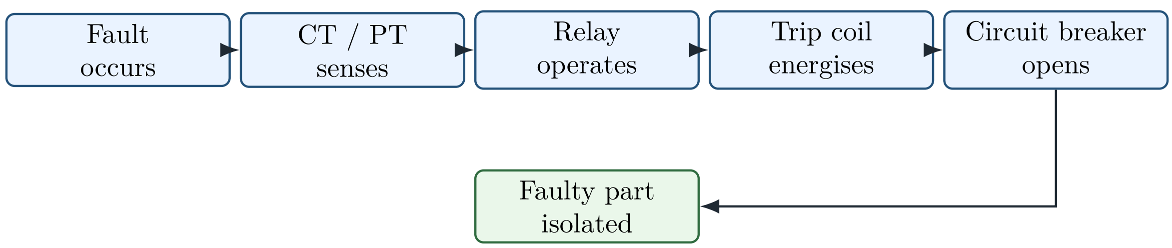
1. protect human life,
2. protect equipment from damage,
3. avoid fire,
4. reduce interruption time,
5. isolate only the faulty section,
6. maintain continuity of supply to healthy parts.

**Memory Hook**

Protection system does two jobs:

Detect fault    Disconnect faulty part

## 3 Basic Protection Chain



**How to Read the Diagram**

| Word/Symbol | Meaning |
|---|---|
| Fault occurs | Abnormal condition like short circuit, earth fault or overload appears. |
| CT/PT senses | Current transformer or potential transformer gives safe signal to relay. |
| Relay operates | Relay detects abnormal current or voltage and closes its trip contact. |
| Trip coil energises | Trip coil of circuit breaker receives current and releases breaker mechanism. |
| Circuit breaker opens | The faulty circuit is disconnected automatically. |
| Faulty part isolated | Only the faulty section is separated from the healthy system. |

## 4 Fuse

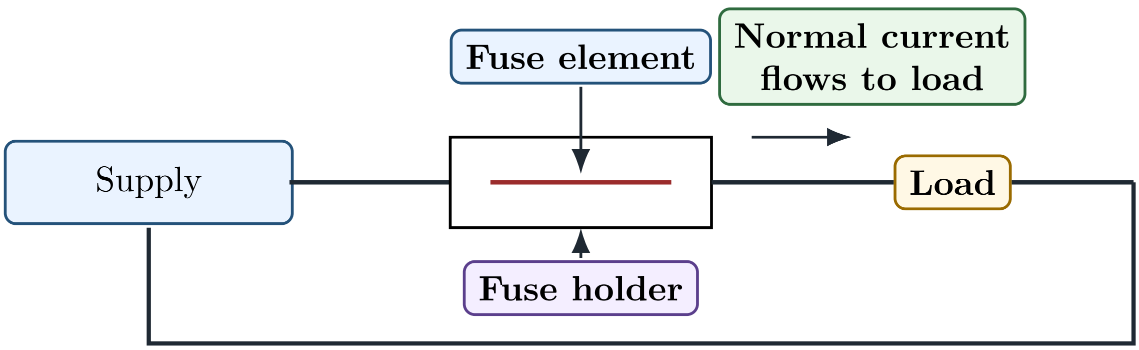
**Definition**

A **fuse** is a protective device having a short piece of metal which melts when excessive current flows through it and disconnects the circuit.

The metal piece inside the fuse is called the fuse element.

Fuse protects the circuit by melting under excessive current.

## 5 Simple Fuse Circuit Diagram



**How to Read the Diagram**

| Word/Symbol | Meaning |
|---|---|
| Supply | Source of electrical energy. |
| Fuse holder | Insulating support which holds the fuse element. |
| Fuse element | Low melting point metal strip which melts during excessive current. |
| Load | Equipment connected to the supply. |
| Normal current | Current flowing safely during normal operation. |
| Return line | Path by which current returns to supply. |

## 6 Working of Fuse

During normal condition:

$$I_{\text{load}} < I_{\text{fusing}}$$

The fuse element does not melt.

During overload or short circuit:

$$I_{\text{fault}} > I_{\text{fusing}}$$

Large heat is produced:

$$H \propto I^2 R t$$

The fuse element melts and opens the circuit.

$$\boxed{\text{Higher current} \Rightarrow \text{more heat} \Rightarrow \text{fuse melts faster}}$$

## 7 Important Fuse Terms

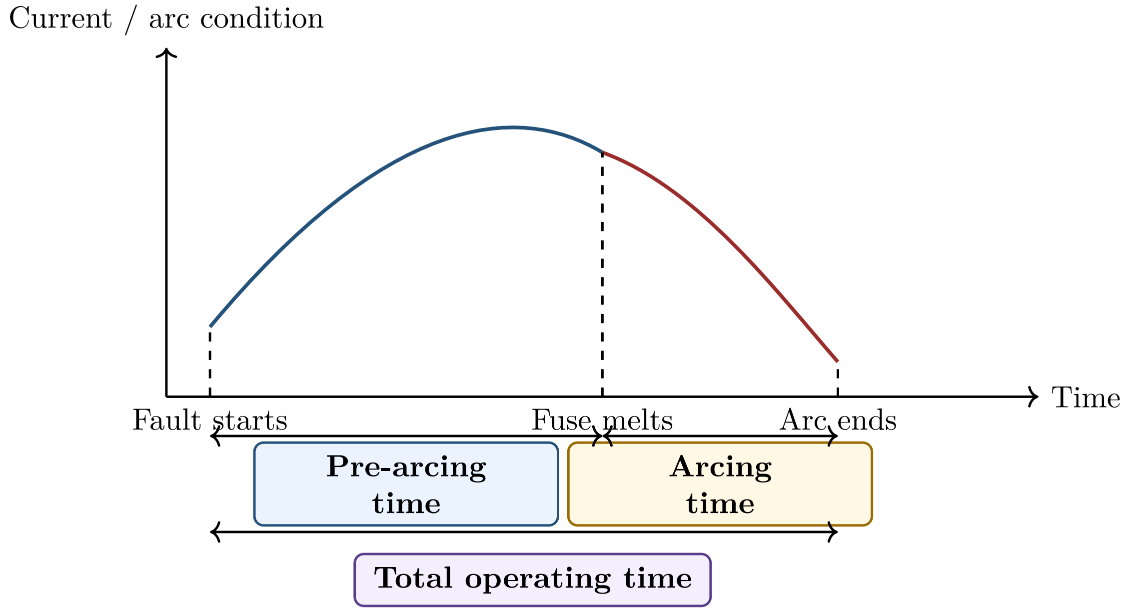
| Term | Meaning |
|---|---|
| Fuse element | The metal part of fuse which melts during excessive current. |
| Current rating | Maximum current which the fuse can carry continuously without melting. |
| Fusing current | Minimum current at which the fuse element melts. |
| Fusing factor | Ratio of fusing current to current rating. |
| Cut-off current | Maximum instantaneous current reached before the fuse interrupts the circuit. |
| Prospective current | Fault current that would flow if the fuse were replaced by a solid conductor of negligible resistance. |
| Pre-arcing time | Time between start of fault and beginning of arc. It is the time taken by fuse element to melt. |

| | |
|---|---|
| Arcing time | Time from beginning of arc to complete arc extinction. |
| Total operating time | Sum of pre-arcing time and arcing time. |
| Breaking capacity | Maximum current that the fuse can safely interrupt at rated voltage. |

$$\boxed{\text{Fusing factor} = \frac{\text{Fusing current}}{\text{Current rating}}}$$

$$\boxed{\text{Total operating time} = \text{Pre-arcing time} + \text{Arcing time}}$$

## 8 Fuse Operation Time Diagram




**How to Read the Diagram**

| Word/Symbol | Meaning |
|---|---|
| Time | Horizontal axis showing time after fault starts. |
| Fault starts | Moment when abnormal current begins. |
| Fuse melts | Moment when fuse element becomes molten and arc begins. |
| Arc ends | Moment when arc is completely extinguished and circuit is interrupted. |
| Pre-arcing time | Time from fault start to melting of fuse element. |
| Arcing time | Time from fuse melting to final arc extinction. |
| Total operating time | Complete time taken by fuse to clear the fault. |

## 9 Desirable Properties of Fuse Element

A good fuse element should have:

1. low melting point,
2. high conductivity,
3. low cost,
4. low deterioration due to oxidation,
5. stable characteristics,
6. suitable mechanical strength,
7. quick operation under fault.

Common fuse element materials are:

- tin,
- lead-tin alloy,
- silver,
- copper.

# 10 Advantages and Disadvantages of Fuse

## 10.1 Advantages

1. Simple and cheap.
2. Requires no maintenance.
3. Operates automatically.
4. Very fast under heavy fault.
5. Can interrupt large short-circuit current if properly rated.

## 10.2 Disadvantages

1. Fuse must be replaced after operation.
2. It cannot be used as a switch for repeated operation.
3. It has limited control flexibility.
4. Exact discrimination may be difficult.
5. It cannot be reset like a circuit breaker.

# 11 HRC Fuse

**Definition**

**HRC fuse** means **High Rupturing Capacity fuse**. It is a fuse designed to safely interrupt very high fault current without explosion or fire.

HRC fuse is used where high short-circuit current may occur.

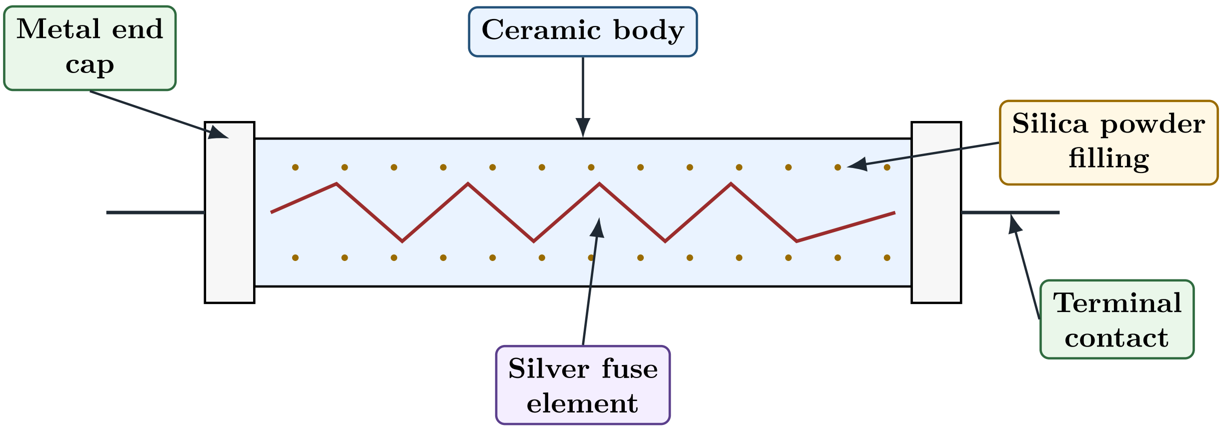
$$\boxed{\text{HRC fuse} = \text{High Rupturing Capacity Fuse}}$$

## 12 Construction of HRC Fuse

An HRC fuse has:

1. ceramic body,
2. silver fuse element,
3. silica powder filling,
4. end caps,
5. terminal contacts,
6. striker pin in some types.

## 13 Clean Diagram of HRC Fuse




Construction of HRC Fuse

**How to Read the Diagram**

| **Word/Symbol** | **Meaning** |
|---|---|
| Ceramic body | Strong insulating body which can withstand heat and pressure. |
| Silica powder filling | Filler material which absorbs heat and helps extinguish arc. |
| Silver fuse element | Conducting element which melts during fault current. Silver gives stable fuse characteristics. |
| Metal end cap | Conducting cap fitted at both ends of the fuse body. |
| Terminal contact | External connection point of the fuse. |
| Construction of HRC Fuse | Shows the internal parts of a cartridge type HRC fuse. |

## 14 Working of HRC Fuse

During normal current, the silver element carries current safely.

During short circuit:

1. Very high current flows.

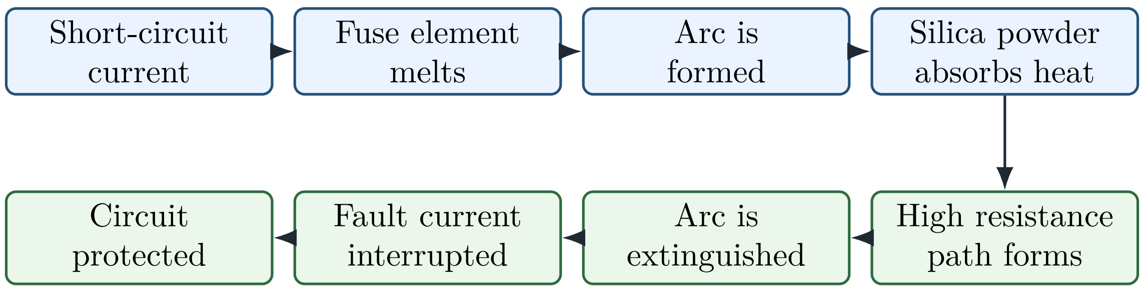
2. Fuse element melts at weak points.
3. Arc is produced.
4. Silica powder absorbs heat.
5. Silica changes into high-resistance glassy material.
6. Arc is extinguished quickly.
7. Faulty circuit is disconnected.

**Memory Hook**

In HRC fuse:

Fuse element melts → Arc forms → Silica absorbs heat → Arc extinguishes

## 15 HRC Fuse Working Flowchart




**How to Read the Diagram**

| Step | Meaning |
|---|---|
| Short-circuit current | Very high current flows due to fault. |
| Fuse element melts | Silver fuse element melts due to $I^2R$ heating. |
| Arc is formed | After melting, current continues for a short time through arc. |
| Silica powder absorbs heat | Silica absorbs arc heat and helps cooling. |
| High resistance path forms | Silica changes into glassy material and increases path resistance. |
| Arc is extinguished | The conducting arc is stopped. |
| Fault current interrupted | Fault current becomes zero. |
| Circuit protected | The faulty circuit is safely disconnected. |

## 16 Advantages of HRC Fuse

1. High breaking capacity.
2. Fast operation during short circuit.
3. No smoke or flame.
4. Reliable and stable operation.

5. Requires less maintenance.
6. Good current-limiting property.
7. Compact size.

### 16.1 Disadvantages of HRC Fuse

1. It must be replaced after operation.
2. It cannot be used for frequent switching.
3. Characteristics cannot be adjusted like relay.
4. Initial cost is higher than simple rewirable fuse.

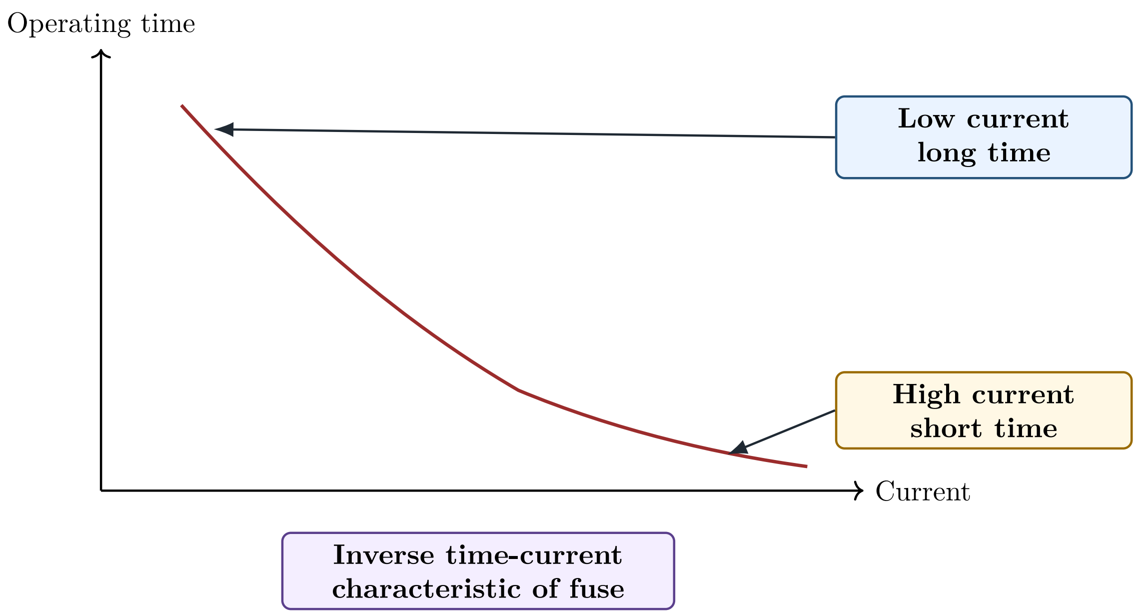
## 17 Time-Current Characteristic of Fuse

**Definition**

The **time-current characteristic** of a fuse shows the relation between current flowing through the fuse and the time taken by the fuse to operate.

Higher current causes faster operation.

$$\boxed{\text{Current increases} \Rightarrow \text{Operating time decreases}}$$



**How to Read the Diagram**

| **Word/Symbol** | **Meaning** |
|---|---|
| Current | Horizontal axis. It shows current flowing through the fuse. |
| Operating time | Vertical axis. It shows time taken by fuse to clear the fault. |
| Low current long time | Slight overload may take longer time to melt the fuse. |
| High current short time | Heavy fault current melts the fuse very quickly. |
| Inverse characteristic | Operating time decreases as current increases. |

## 18 Protective Relay

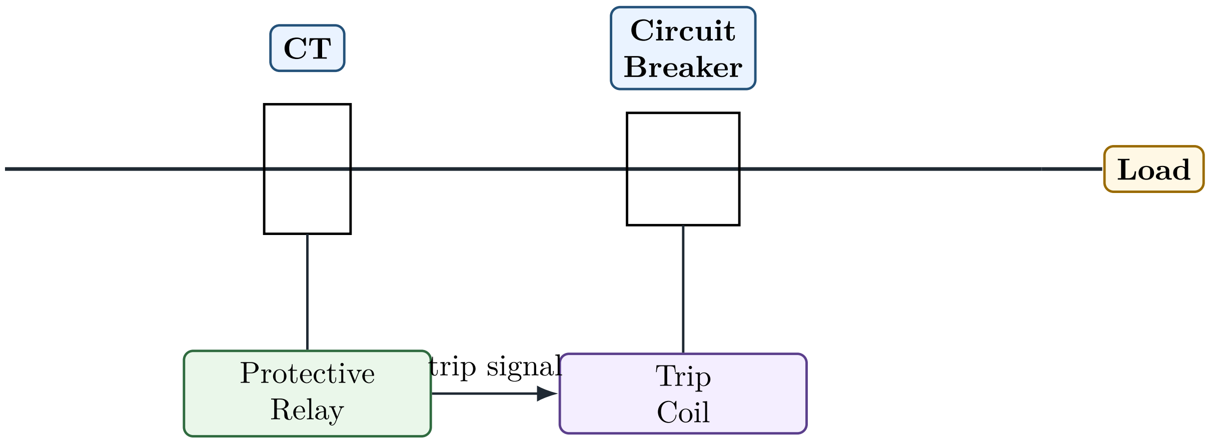
**Definition**

A **protective relay** is a device that detects abnormal conditions in an electrical circuit and causes the circuit breaker to trip.

Relay does not usually interrupt the main fault current directly. It senses fault and sends trip command to the circuit breaker.

Relay detects fault; circuit breaker clears fault.

## 19 Relay Protection Diagram




**Relay detects abnormal current and trips circuit breaker**

### How to Read the Diagram

| Word/Symbol | Meaning |
|---|---|
| Power line | Main circuit carrying load current. |
| CT | Current transformer. It gives reduced current signal to relay. |
| Protective Relay | Device which senses abnormal current and gives trip signal. |
| Trip signal | Electrical command sent from relay to trip coil. |
| Trip Coil | Coil which operates the circuit breaker tripping mechanism. |
| Circuit Breaker | Device which opens the main circuit and interrupts fault current. |
| Load | Consumer or equipment connected to the protected circuit. |

## 20 Essential Qualities of Protective Relay

A good relay protection system should have:

1. Reliability,
2. Selectivity,
3. Sensitivity,
4. Speed,
5. Stability,
6. Simplicity,
7. Economy.

| Quality | Meaning |
|---|---|
| Reliability | Relay must operate when it should and must not operate when it should not. |
| Selectivity | Only the faulty section should be disconnected. |
| Sensitivity | Relay should detect even small abnormal conditions. |
| Speed | Fault should be cleared quickly. |
| Stability | Relay should not operate for external faults or normal conditions. |
| Simplicity | Scheme should be easy to understand and maintain. |
| Economy | Protection cost should be reasonable. |

## 21 Circuit Breaker

### Definition

A **circuit breaker** is a switching device which can make, carry and break current under normal as well as abnormal fault conditions.

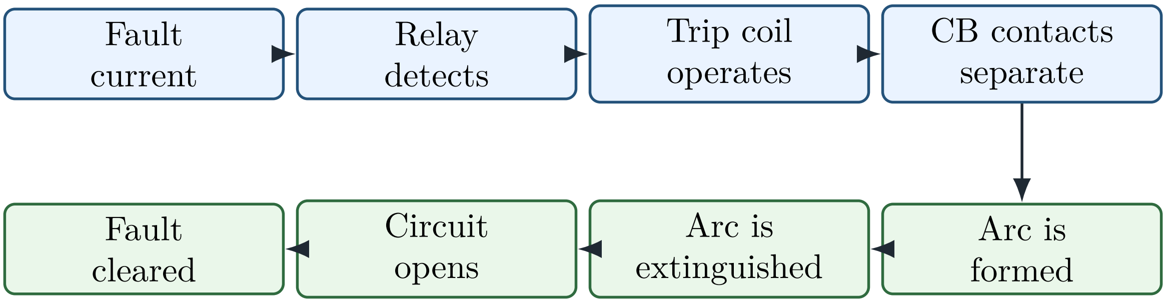
A circuit breaker can be opened:

- manually,

- automatically by relay operation.

$\boxed{\text{Circuit breaker interrupts fault current.}}$

## 22 Circuit Breaker Working Sequence




**How to Read the Diagram**

| Step | Meaning |
|---|---|
| Fault current | Abnormally high current flows in the circuit. |
| Relay detects | Relay senses the abnormal condition through CT or PT. |
| Trip coil operates | Trip coil receives signal and releases breaker mechanism. |
| CB contacts separate | Main contacts of circuit breaker open. |
| Arc is formed | Current continues momentarily through arc between separating contacts. |
| Arc is extinguished | Breaker medium extinguishes the arc. |
| Circuit opens | Current path becomes open. |
| Fault cleared | Faulty part is disconnected from the system. |

## 23 Types of Circuit Breakers

Common circuit breakers are:

1. Oil circuit breaker,
2. Air blast circuit breaker,
3. $SF_6$ circuit breaker,
4. Vacuum circuit breaker,
5. Miniature circuit breaker,
6. Moulded case circuit breaker.

| Type | Main Idea |
|---|---|
| Oil circuit breaker | Oil is used for insulation and arc extinction. |
| Air blast circuit breaker | High-pressure air blast extinguishes arc. |
| $SF_6$ circuit breaker | Sulphur hexafluoride gas is used for arc extinction. |

| | |
|---|---|
| Vacuum circuit breaker | Contacts open in vacuum; arc is quickly extinguished. |
| MCB | Low-voltage automatic protective switch for small circuits. |
| MCCB | Low-voltage breaker for higher current circuits than MCB. |

# 24 Isolator

**Definition**

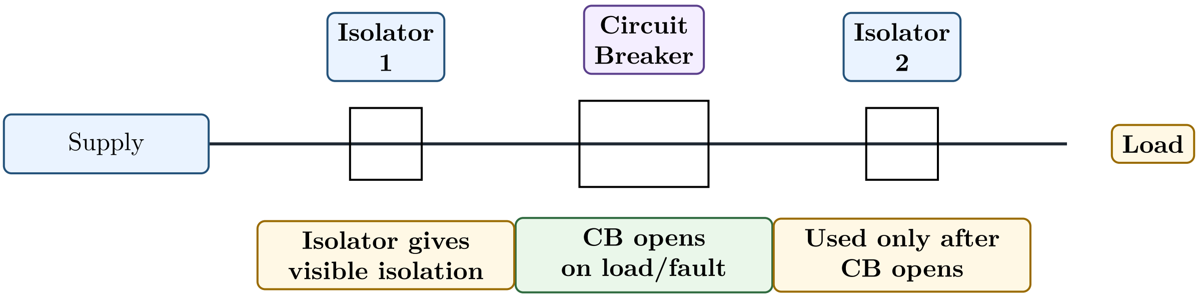
An **isolator** is a mechanical switch used to isolate a part of the circuit after the circuit has been made dead.

Important point:

Isolator should not be opened on load.

It is used only after circuit breaker has opened the circuit.

# 25 Circuit Breaker and Isolator Arrangement




**How to Read the Diagram**

| Word/Symbol | Meaning |
|---|---|
| Supply | Source side of the circuit. |
| Load | Equipment receiving power. |
| Isolator 1 and Isolator 2 | Mechanical switches used for visible isolation during maintenance. |
| Circuit Breaker | Device capable of interrupting load current and fault current. |
| CB opens on load/fault | Circuit breaker can safely open current-carrying circuit. |
| Visible isolation | Isolator gives clear physical separation after circuit is dead. |
| Used only after CB opens | Isolator must not be opened while current is flowing. |

# 26 Correct Operating Sequence

## 26.1 To Shut Down a Circuit

Open circuit breaker first

Then open isolators

### 26.2 To Energise a Circuit

Close isolators first

Then close circuit breaker

**Memory Hook**

Remember:

CB handles current. Isolator handles isolation.

## 27 Comparison: Fuse and Circuit Breaker

| **Point** | **Fuse** | **Circuit Breaker** |
|---|---|---|
| Function | Protects by melting | Protects by opening contacts |
| Operation | Automatic only | Manual and automatic |
| Replacement | Must be replaced after operation | Can be reset after operation |
| Cost | Cheaper | Costlier |
| Maintenance | Very low | Requires maintenance |
| Control | No remote control generally | Remote and automatic control possible |
| Use | Low and medium protection applications | Repeated switching and fault clearing |

## 28 Comparison: Relay and Circuit Breaker

| **Point** | **Relay** | **Circuit Breaker** |
|---|---|---|
| Main job | Detects fault | Interrupts fault current |
| Current handling | Handles small signal current | Handles main circuit current |
| Connection | Connected through CT/PT | Connected in main power circuit |
| Output | Sends trip command | Opens circuit contacts |
| Can clear fault alone? | No | Yes, after trip command |

## 29 Comparison: Circuit Breaker and Isolator

| **Point** | **Circuit Breaker** | **Isolator** |
|---|---|---|
| Purpose | Interrupts current | Provides isolation |
| Operation on load | Can operate on load | Should not operate on load |

| | | |
|---|---|---|
| Arc extinction | Has arc extinction system | No arc extinction system |
| Automatic operation | Possible through relay | Generally manual |
| Use | Protection and switching | Maintenance isolation |
| Operating sequence | Open first during shutdown | Open after circuit breaker |

# 30 Important Small Numericals

## 30.1 Numerical 1: Fusing Factor

**Problem:** A fuse has current rating of 20 A and fusing current of 30 A. Find fusing factor.

**Solution:**

$$\text{Fusing factor} = \frac{\text{Fusing current}}{\text{Current rating}}$$

$$= \frac{30}{20}$$

$$\boxed{\text{Fusing factor} = 1.5}$$

## 30.2 Numerical 2: Total Operating Time

**Problem:** A fuse has pre-arcing time 0.02 s and arcing time 0.01 s. Find total operating time.

**Solution:**

$$\text{Total operating time} = \text{Pre-arcing time} + \text{Arcing time}$$

$$= 0.02 + 0.01$$

$$\boxed{\text{Total operating time} = 0.03 \text{ s}}$$

## 30.3 Numerical 3: Prospective Current

**Problem:** A supply voltage is 250 V and total fault path impedance is $0.05\Omega$. Find prospective current.

**Solution:**

$$I_{\text{prospective}} = \frac{V}{Z}$$

$$I_{\text{prospective}} = \frac{250}{0.05}$$

$$\boxed{I_{\text{prospective}} = 5000 \text{ A}}$$

**Semester Answer Style**

Prospective current means the fault current that would flow if the protective device did not interrupt the circuit.

## 31 Reverse Engineering of Semester Questions

| Syllabus Word | Concept | Semester Question Form |
|---|---|---|
| Fuse | Melting protective device | Define fuse and explain working with diagram. |
| Fuse terms | Rated current, fusing current, fusing factor | Define prospective current, pre-arcing time, arcing time. |
| HRC fuse | High rupturing capacity fuse | Draw and explain construction and working of HRC fuse. |
| Protection | Detect and isolate fault | Draw protection flow using relay and circuit breaker. |
| Protective relay | Fault detecting device | Define relay and list qualities of protective relay. |
| Circuit breaker | Fault current interrupting device | Explain working sequence of circuit breaker. |
| Isolator | Off-load mechanical isolation switch | Difference between circuit breaker and isolator. |
| Comparison | Fuse vs CB, relay vs CB | Write comparison tables. |

## 32 Most Predictable Semester Questions from Phase 8

| No. | Question | Marks |
|---|---|---|
| 1 | Define fuse. Explain its working with a neat diagram. | 4–5 |
| 2 | Define current rating, fusing current and fusing factor. | 3–4 |
| 3 | Define prospective current, pre-arcing time, arcing time and breaking capacity. | 4–6 |
| 4 | Draw and explain construction of HRC fuse. | 5–6 |
| 5 | Explain working of HRC fuse. | 5 |
| 6 | Draw time-current characteristic of fuse and explain. | 4 |
| 7 | What is protective relay? Mention essential qualities of protective relay. | 5 |
| 8 | Draw basic relay-circuit breaker protection scheme and explain. | 5–6 |
| 9 | Define circuit breaker. Explain its working sequence. | 4–5 |
| 10 | Differentiate between fuse and circuit breaker. | 4–6 |
| 11 | Differentiate between circuit breaker and isolator. | 4–6 |
| 12 | Explain correct operating sequence of circuit breaker and isolator. | 3–4 |

## 33 Exam Answer Templates

### 33.1 Template 1: Fuse Answer

**Semester Answer Style**

**Answer sequence:**

1. Define fuse.
2. Draw simple fuse circuit diagram.
3. Label fuse element, fuse holder, supply and load.
4. Explain normal operation.
5. Explain fault operation using $I^2R$ heating.
6. Write advantages and disadvantages.

### 33.2 Template 2: HRC Fuse Answer

**Semester Answer Style**

**Answer sequence:**

1. Write full form: High Rupturing Capacity.
2. Draw construction diagram.
3. Label ceramic body, silver element, silica powder, end caps and terminals.
4. Explain working during fault.
5. Write advantages.

### 33.3 Template 3: Relay and Circuit Breaker Answer

**Semester Answer Style**

**Answer sequence:**

1. Define protective relay.
2. Define circuit breaker.
3. Draw CT–Relay–Trip Coil–CB diagram.
4. Explain that relay detects fault.
5. Explain that circuit breaker interrupts fault current.
6. Write qualities of relay or comparison if asked.

### 33.4 Template 4: Isolator Answer

**Semester Answer Style**

**Answer sequence:**

1. Define isolator.
2. State that isolator should not be opened on load.
3. Draw CB and isolator arrangement.
4. Write shutdown sequence:

   Open CB first, then open isolator.

5. Write energising sequence:

   Close isolator first, then close CB.

## 34 Phase 8 Final Memory Sheet

**One-Page Memory Sheet**

1. Protection detects faults and disconnects faulty parts.
2. Fuse protects by melting during excessive current.
3. Heating in fuse:
$$H \propto I^2 Rt$$
4. Fusing factor:
$$\text{Fusing factor} = \frac{\text{Fusing current}}{\text{Current rating}}$$
5. Total operating time:
$$t_{\text{total}} = t_{\text{pre-arcing}} + t_{\text{arcing}}$$
6. Prospective current is the fault current that would flow if fuse did not interrupt.
7. HRC fuse means High Rupturing Capacity fuse.
8. In HRC fuse, silica powder helps arc extinction.
9. Relay detects fault and gives trip command.
10. Circuit breaker interrupts fault current.
11. Isolator gives visible isolation and should not be opened on load.
12. Shutdown sequence:
$$\boxed{\text{Open CB first, then open isolator}}$$
13. Energising sequence:
$$\boxed{\text{Close isolator first, then close CB}}$$

## 35 Phase 8 Self-Test

Answer these without seeing notes:

1. What is protection in power system?

2. Why is protection necessary?
3. What is a fuse?
4. What is fuse element?
5. Define current rating and fusing current.
6. What is fusing factor?
7. What is prospective current?
8. What is pre-arcing time?
9. What is arcing time?
10. What is HRC fuse?
11. Why is silica powder used in HRC fuse?
12. What is a protective relay?
13. What is the job of a circuit breaker?
14. What is an isolator?
15. Why should isolator not be opened on load?
16. Compare fuse and circuit breaker.
17. Compare circuit breaker and isolator.

**Common Mistake**

Do not write that relay directly breaks the main circuit. Correct statement:

Relay detects fault and circuit breaker interrupts fault current.

## End of Phase 8

**Next Phase:**
**Phase 9 – Substation and Busbar Arrangement**

Substation classification, equipment, CT, PT, isolator, circuit breaker, lightning arrester, pole-mounted substation and busbar schemes.